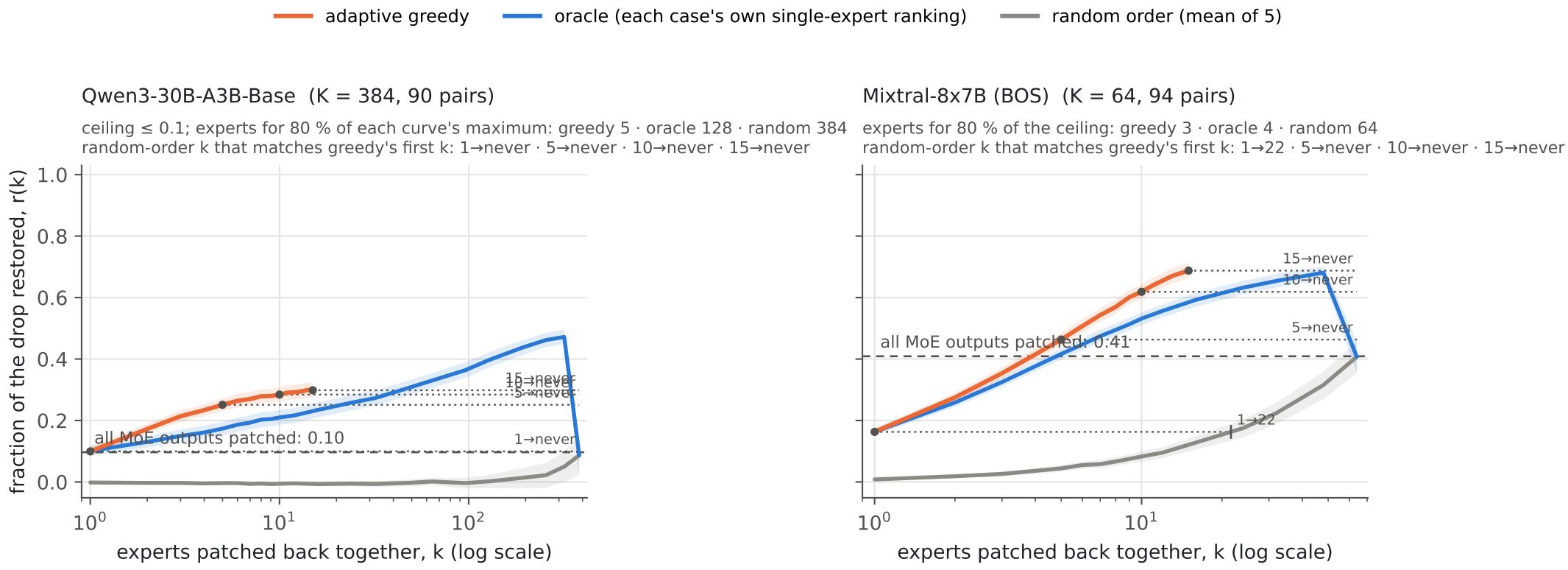direction: F1 on the Phase-3 twins: add-back curves (validation)